DLBDSEDA02 - Project: Data Analysis
Phase 2: Development and Reflection

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During Phase 2, the selected CFPB complaint sample was validated and the text analysis pipeline was implemented reproducibly. The file contained 5,000 complaints, but 683 exact duplicate narratives were removed and one additional record became empty after preprocessing, leaving 4,316 documents for modeling. A key discovery was that the sample is not a cross-section of all CFPB complaints: every record concerns incorrect information on a credit report. The project scope was therefore refined to recurring subthemes within this specific complaint category.

Two vectorization techniques were compared: Bag of Words and TF-IDF. LDA was applied to the Bag-of-Words matrix and LSA/TruncatedSVD to TF-IDF. Candidate topic counts from four to eight were tested using NPMI coherence and topic diversity, together with LDA perplexity and LSA explained variance. The executed selection produced an eight-topic LDA solution and a four-component LSA solution. LDA gave more granular themes and probabilistic topic prevalence, while LSA produced broader latent dimensions.

One implementation issue was the planned WordNet lemmatization, because the execution environment could not retrieve NLTK corpora. The final preprocessing was changed to a deterministic offline procedure using scikit-learn stop words. A further limitation is repeated legal/template language, which can influence the discovered topics even after exact duplicates are removed.

GitHub repository: https://github.com/arriess/consumer-complaint-topic-analysis